

Question	Marking Guidance	Mark	Comments
06.1	1. (In) DNA; 2. (In) RNA; 3. (In) ATP/ADP; 4. Phosphorylation; 5. (In) phospholipids;	2 max (2 x AO1)	1. and 2. If neither DNA or RNA named allow 1 mark for nucleotide/nucleic acid/phosphodiester bond/sugar-phosphate backbone 1, 2 and 3. Accept any other correct biological compound containing phosphate; eg (in)RuBP or GP or triose phosphate or NAD 4. Accept binds with substance to make it more reactive

Question	Marking Guidance	Mark	Comments
06.2	1. Increased (plasma) acidity OR Decreased (plasma) pH; 2. Denatures (protein) OR Changes (protein) tertiary structure OR Changes active site (shape) OR Changes antigen-binding site (shape);	2 (2 x AO1)	2. Accept description of bond changes, eg 'disrupts hydrogen/ionic bonds' for 'changes in tertiary structure'

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06.3	1. Equal positive and negative (in both); 2. Higher (ion concentration) in cytoplasm;	2 (2 x AO2)	Accept answers in either order Ignore reference to sodium ions 1. Accept no overall charge or no net charge 2. Ignore more ions in cytoplasm

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06.4	1. (Sodium) ions move in (to cells) by <u>facilitated</u> diffusion down a concentration gradient; 2. (Sodium) ions move out (of cells) by active transport against a concentration gradient;	2 (2 x AO2)	1. Accept a description of the concentration gradient, eg from high (concentration) to low (concentration) 2. Accept a description of transport against a gradient, eg from low (concentration) to high (concentration)